

Chapter 3

The Edentulous Ridge

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Clinical considerations

The alveolar process extends from the basal bone of the maxilla or the mandible and forms a boundary between the outer portion of the maxilla and the inner portion of the mandible (Pietrokovski *et al.* 2007). The alveolar process forms in harmony with the development and eruption of the teeth, and gradually regresses when the teeth are lost. In other words, the formation as well as the preservation of the alveolar process is dependent on the continued presence of teeth. Furthermore, the morphologic characteristics of the alveolar process are related to the size and shape of the teeth, events occurring during tooth eruption, as well as the inclination of the erupted teeth. Thus, subjects with long and narrow teeth, compared with subjects who have short and wide teeth, appear to have a more delicate alveolar process and, in particular in the front tooth regions, a thin, sometimes fenestrated, buccal bone plate (Fig. 3-1).

The tooth and its surrounding attachment tissues – the root cementum, the periodontal ligament, and the bundle bone – establish a functional unit (Fig. 3-2). Hence, forces elicited, for example during mastication, are transmitted from the crown of the tooth via the root and the attachment tissues to the load-carrying hard tissue structures in the alveolar process, where they are dispersed.

The loss of teeth and the loss or change of function within and around the socket will result in a series of adaptive alterations of the now edentulous portion of

the ridge. Thus, it is well documented that following *multiple tooth* extractions and the subsequent restoration with removable dentures, the size of the ridge will become markedly reduced, not only in the horizontal but also in the vertical dimension (Figs. 3-3, 3-4). An important long-term study of dimensional ridge alterations in 42 complete denture wearers was presented by Bergman & Carlsson (1985). Cephalometric radiographic examinations were performed in a cephalostat and profiles of the edentulous mandible and maxilla were depicted 2 days after tooth extraction, and subsequently after 5 years and 21 years (Fig. 3-5). The authors concluded that during the observation interval most of the hard tissue component of the ridge was lost. However, there was wide variation in the degree of bone resorption and amount of remaining bone among the patients (Tallgren 1957, 1966; Atwood 1962, 1963; Johnson 1963, 1969; Carlsson *et al.* 1967).

Also, following the removal of *single* teeth, the ridge at the site will be markedly diminished (Fig. 3-6). The magnitude of this change was studied and reported by Pietrokovski and Massler (1967). The authors had access to 149 dental cast models (72 maxillary and 77 mandibular) in which one tooth was missing on one side of the jaw. The outer contours of the buccal and lingual (palatal) portions of the ridge at a tooth site and at the contralateral edentulous site were determined by the use of a profile stylus and an imaging technique. Their findings are reported in Table 3-1.

It was concluded that the amount of tissue resorption (hard and soft tissues combined) following the loss

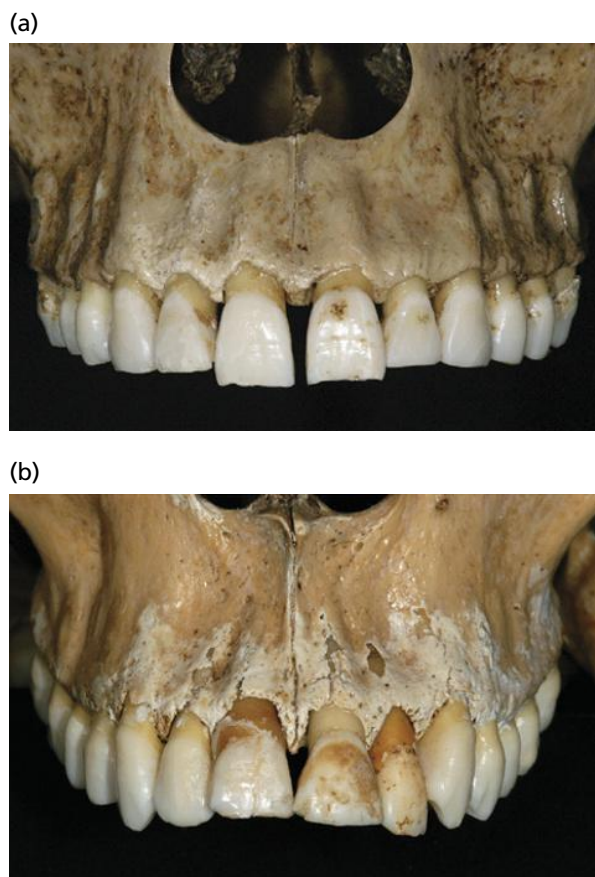


Fig. 3-1 Buccal aspect of adult skull preparations illustrating a dentate maxilla of one subject with a relatively thick (a) and another subject with a relatively thin (b) biotype.

of a single tooth was substantial and that the reduction of the ridge was twice as large at the buccal aspect as along the lingual and palatal aspect in all teeth groups examined. The absolute amounts of tissue loss varied from one group of teeth to the next. As a result of this tissue modeling, the center of the edentulous site shifted toward the lingual or palatal aspect of the ridge. The observations made by Pietrokovski and Massler (1967) were supported by findings presented by Schropp *et al.* (2003). They studied bone and soft tissue volume changes that took place during a 12-month period following the extraction of single premolars and molars. Clinical as well as cast model measurements were made immediately after tooth extraction and subsequently after 3, 6, and 12 months of healing. It was observed that the bucco-lingual/-palatal dimension during the first 3 months was reduced by about 30%, and after 12 months the edentulous site had lost at least 50% of its original width. Furthermore, the height of the buccal bone plate was reduced and after 12 months of healing the buccal prominence was located 1.2mm apical of its lingual/palatal counterpart.

The information provided by Pietrokovski and Massler (1967) and Schropp *et al.* (2003) suggests that if an alveolar process includes a tooth that has a horizontal width of, for example, 12mm, the edentulous site will be only 6mm wide 12 months after healing following tooth extraction. During this 12-month interval, 4mm of tissue will be lost from the buccal and 2mm from the lingual aspect of the site.

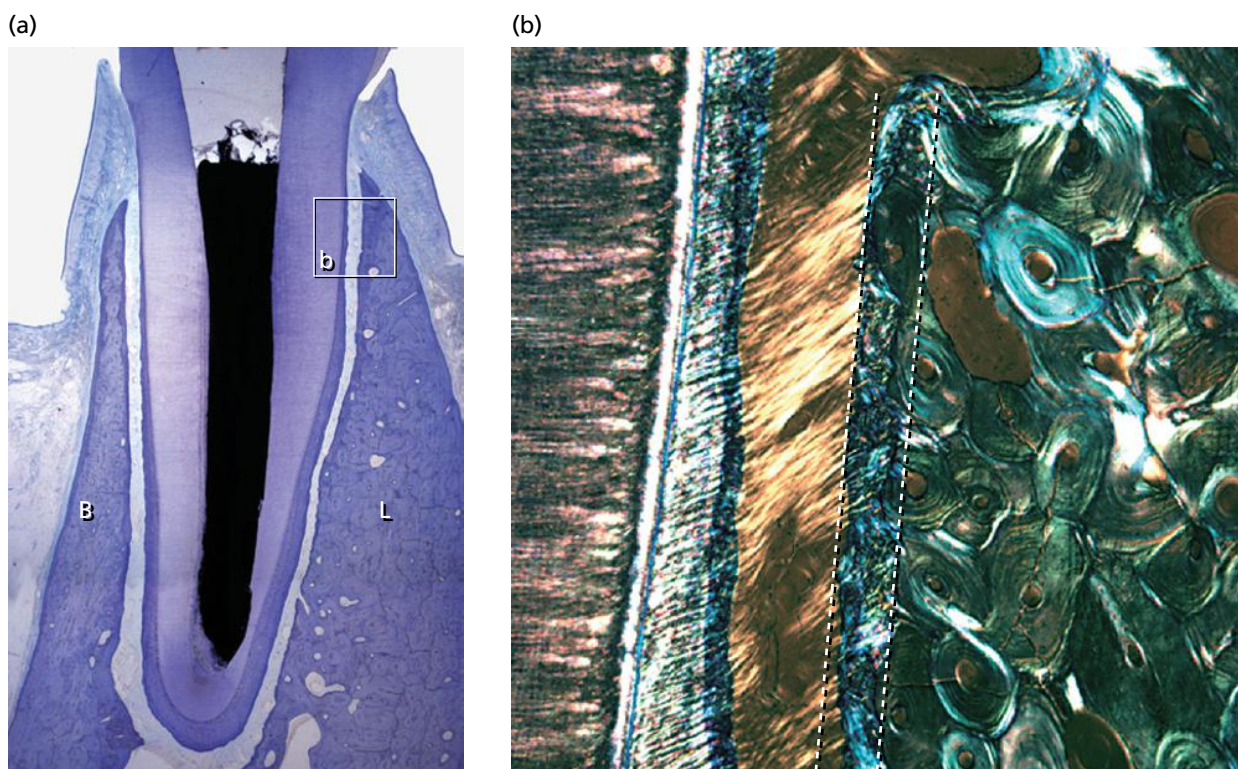


Fig. 3-2 Buccolingual histologic section of the alveolar process. (a) Tooth is surrounded by its attachment tissues (cementum, periodontal ligament, alveolar bone proper). (B, buccal aspect; L, lateral aspect.) (b) Higher magnification of the attachment tissues. Note that the dentin is connected to the alveolar bone via the root cementum, and the periodontal ligament. The alveolar bone is characterized by its content of circumferential lamellae. The portion of the bone that is facing the periodontal ligament (between the dotted lines) is called the alveolar bone proper or the bundle bone.

(a)

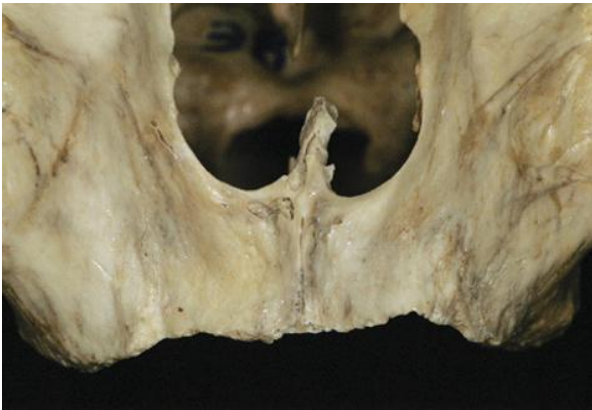


(b)



Fig. 3-3 (a) Clinical view of a partially edentulous maxilla. Note that the crest of the edentulous portions of the ridge is narrow in the buccopalatal direction. (b) Clinical view of a fully edentulous and markedly resorbed maxilla. Note that *papilla incisivae* is located in the center of the ridge. This indicates that the entire buccal and also a substantial portion of the palatal ridge are missing.

(a)



(b)



Fig. 3-4 Buccal aspect of a skull preparation illustrating a fully edentulous maxilla (a) and mandible (b). The small segments of the alveolar ridge that still remain are extremely thin in the bucco-palatal/-lingual direction.

In a clinical study (Sanz *et al.* 2010; Tomasi *et al.* 2010) it was observed that the degree of early (4 months) resorption of the buccal bone plate following tooth extraction was dependent on its original dimension. Thus, bone plates that were <1 mm wide lost substantially more dimension (width and height) than plates that were >1 mm wide.

In this context it is important to acknowledge that the buccal bone plate in the frontal tooth region in humans is frequently (>80% of sites) <1 mm wide (Januário *et al.* 2011; Braut *et al.* 2011; Nowzari *et al.* 2012). Hence, it can be anticipated that tooth loss in this part of the dentition may result in marked dimension alterations (horizontal as well as vertical) of the ridge and that this in turn may cause esthetic concerns.

Conclusion: The extraction of single as well as multiple teeth induces a series of adaptive changes in the soft and hard tissues that result in an overall regression of the edentulous site(s). Resorption appears to be more pronounced at the buccal than at the lingual/palatal aspects of the ridge.

It should be realized that the alveolar process might also undergo change as the result of tooth-related disease processes, such as aggressive, chronic,

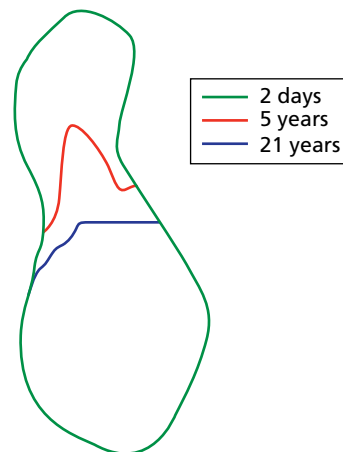


Fig. 3-5 Profile of the mandibular bone following tooth extraction at 2 days, 5 years, and 21 years after tooth removal. (Source: Bergman & Carlsson 1985. Reproduced with permission from Elsevier.)

and necrotizing forms of marginal periodontitis, as well as periapical periodontitis. Furthermore, traumatic injuries (including from improper tooth removal techniques) may cause marked damage to the alveolar process of the maxilla and mandible.

(a)



(b)



Fig. 3-6 Clinical view of an edentulous ridge in the maxillary premolar region. The premolar was extracted several years before the clinical documentation was made. (a) Note the presence of a buccal invagination of the ridge. (b) Following flap elevation, the crest region of the severely resorbed buccal portion of the alveolar process is disclosed.

Table. 3-1 Average amount of resorption of tooth extraction in different tooth areas.^a

Tooth	Average amount of resorption (mm)		Difference
	Buccal surface	Lingual/palatal surface	
Mandibular teeth			
Central incisor	2.08	0.91	1.17
Lateral incisor	3.54	1.41	2.13
Canine	3.25	1.59	1.66
First premolar	3.45	1.40	2.05
Second premolar	3.28	0.75	2.53
First molar	4.69	2.79	1.90
Second molar	4.30	3.00	1.30
Maxillary teeth			
Central incisor	3.03	1.46	1.57
Lateral incisor	3.47	0.86	2.61
Canine	3.33	1.91	1.42
First premolar	3.33	2.04	1.29
Second premolar	2.58	1.62	0.96
First molar	5.25	3.12	2.13

^a "The amount of resorption was greater along the buccal surface than along the lingual or palatal surface in every specimen examined, although the absolute amounts and differences varied very widely. This caused a shift in the center of the edentulous ridge toward the lingual or palatal side of the ridge with a concomitant decrease in arch length in the mandible as well as the maxillae" (Petrokovski & Massler 1967).

Remaining bone in the edentulous ridge

In the publication by Schropp *et al.* (2003), bone tissue formation in single extraction sockets was studied by means of subtraction radiography. Thus, radiographs of the study sites were obtained using a standardized technique immediately after tooth extraction and then after 3, 6, and 12 months

of healing (Fig. 3-7). It was observed that in the first few months, some bone loss (height) took place in the alveolar crest region. Most of the bone gain in the socket occurred in the first 3 months. There was additional gain of bone in the socket between 3 and 6 months. In the interval between 6 and 12 months, the newly formed bone obviously remodeled and the amount of mineralized tissue was reduced. In other words, in the later phases of socket healing, small amounts of mineralized tissue may have remained in the center of the edentulous site.

The bony part of the edentulous ridge in humans was examined in biopsies sampled from the posterior portions of the jaw by Lindhe *et al.* (2012). The peripheral borders of the ridge were consistently lined with dense cortical bone. More central parts harbored cancellous bone and included trabeculae made up mainly of lamellar bone (Fig. 3-8a). The trabeculae that were embedded in bone marrow varied in shape, and frequently had a haphazard orientation. The bone marrow was dominated by adipocytes, vascular structures, and scattered inflammatory cells. The hard tissue component of the ridge was comprised of a mixture of mineralized bone (about 60%), bone marrow (about 20%), and fibrous tissue (about 15%) (Fig. 3-8b).

Classification of remaining bone

Based on the volume of remaining mineralized bone, the edentulous sites may, according to Lekholm and Zarb (1985), be classified into five different groups (Fig. 3-9). In groups A and B, substantial amounts of the ridge still remain, whereas in groups C, D, and E, only minute amounts of hard tissue remain. Lekholm and Zarb (1985) also classified the "quality" of the bone in the edentulous site. Class 1 and class 2 characterized a location in which the walls – the cortical plates – of the site are thick and the volume of bone

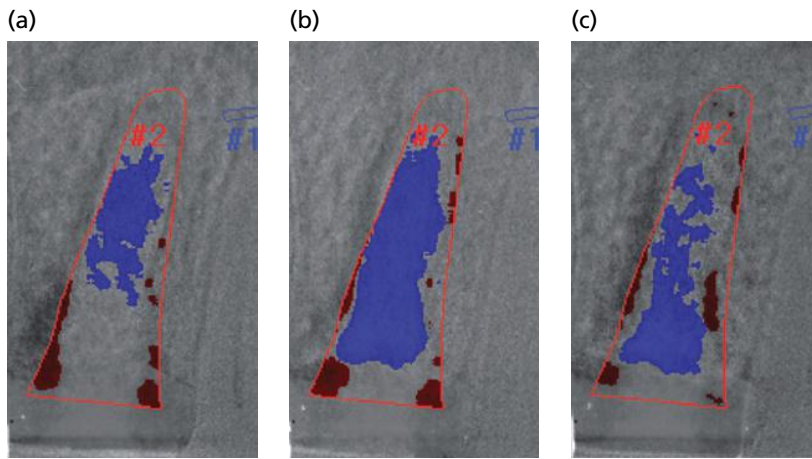


Fig. 3-7 Radiographic (subtraction radiography) images of an extraction site obtained after (a) 3 months, (b) 6 months, and (c) 12 months of healing. The blue color represents areas of new bone formation. During the first 6 months, the deposition of new bone was intense. Between 6 and 12 months, some of the newly formed bone was remodeled. (Courtesy of L. Schropp.)

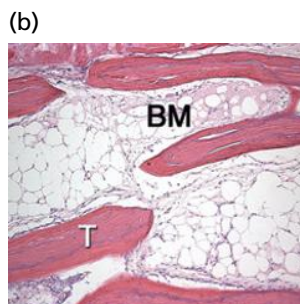
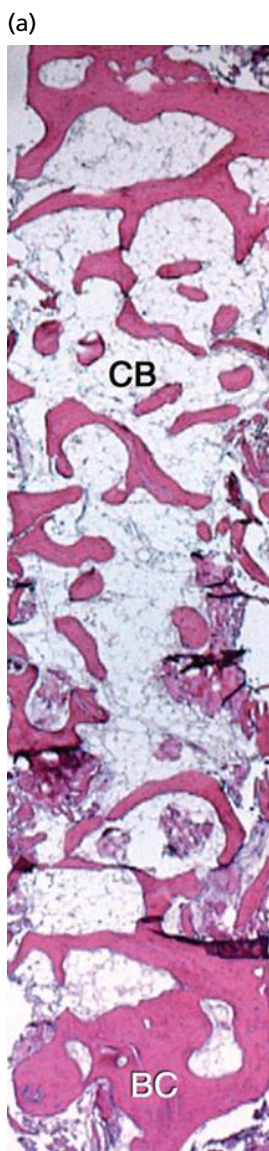


Fig. 3-8 Histologic sections of an edentulous site obtained from the maxillary premolar region in man. (a) The marginal portion of the ridge (BC) is protected by a cortical cap made up of lamellar bone, while more central regions house the cancellous bone (CB). (b) The cancellous bone is characterized by the trabeculae of mineralized bone (T) within the bone marrow (BM) compartment.

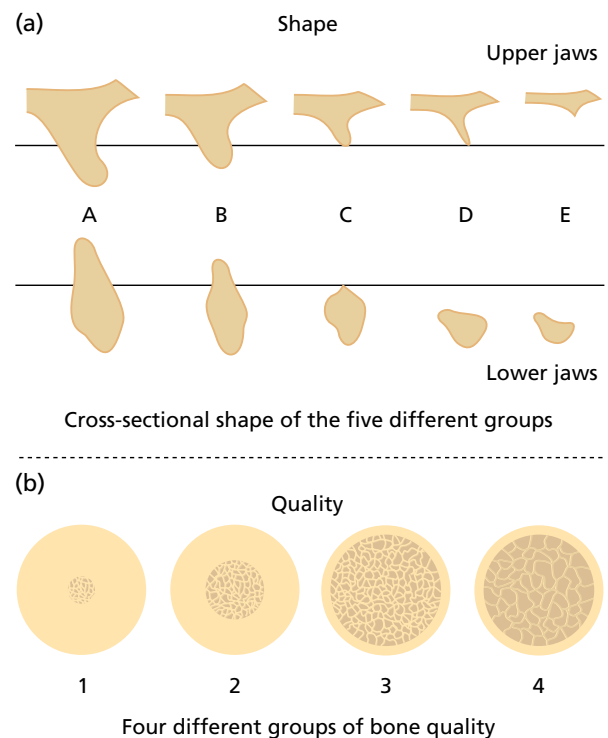


Fig. 3-9 Schematic drawings showing (a) a classification of residual jaw shape and (b) jaw bone quality. (Source: Lekholm & Zarb 1985. Reproduced from Quintessence.)

marrow is small. Relatively thin walls of cortical bone, however, will border sites that belong to class 3 and class 4, while the amount of cancellous bone (spongiosa), including trabeculae of lamellar bone and marrow, is large.

Topography of the alveolar process

The alveolar process that houses the roots of the teeth extends from the basal bone (Fig. 3-10a) of the maxilla and the mandible. The shape and dimensions (height and width) of the basal bone vary considerably from subject to subject (Figs. 3-10a, b) and from site to site in the same individual. There

is no distinct boundary between the alveolar process and the basal bone of the jaws.

At sites of the jaws where the teeth erupt in “normal” orientation in the developing alveolar process, hard tissue will be present on the facial (buccal) as well as on the lingual (palatal) aspect of the roots (Fig. 3-10c). However, at sites where the teeth erupt with a facial orientation, the facial (buccal) bone of the alveolar process will become thin and at times even disappear (dehiscence, fenestration) (Fig. 3-10d).

The outer walls of the alveolar process – facial (buccal), marginal, and lingual (palatal) aspects – are continuous with the outer walls of the basal bone. The walls are comprised of dense cortical bone, while more central portions harbor trabecular bone (radiographic term; spongy bone, anatomic term; cancellous bone, histologic term) that contains bone trabeculae within the bone marrow.

The cortical walls (plates) of the alveolar process are continuous with the bone that lines the sockets, that is the alveolar bone proper or the bundle bone (see Fig. 3-2b). The cortical plates (the outer walls) of the

alveolar process meet the alveolar bone proper at the crest of the interdental septum. In subjects (sites) with healthy periodontium, the crest of the septum is located 1–2 mm apical of the cemento-enamel junction.

In some portions of the dentition (such as in the symphysis region of the mandible), the trabecular bone component of the alveolar process may be absent.

From an alveolar process to an edentulous ridge

The alterations that occur in the alveolar process following the extraction of a single tooth can, for didactic reasons, be divided in two interrelated series of events, namely *intra-alveolar processes* and *extra-alveolar processes*.

Intra-alveolar processes

The healing of extraction sockets in human volunteers was studied by, for example, Amler (1969) and Evian *et al.* (1982). Although the biopsy technique

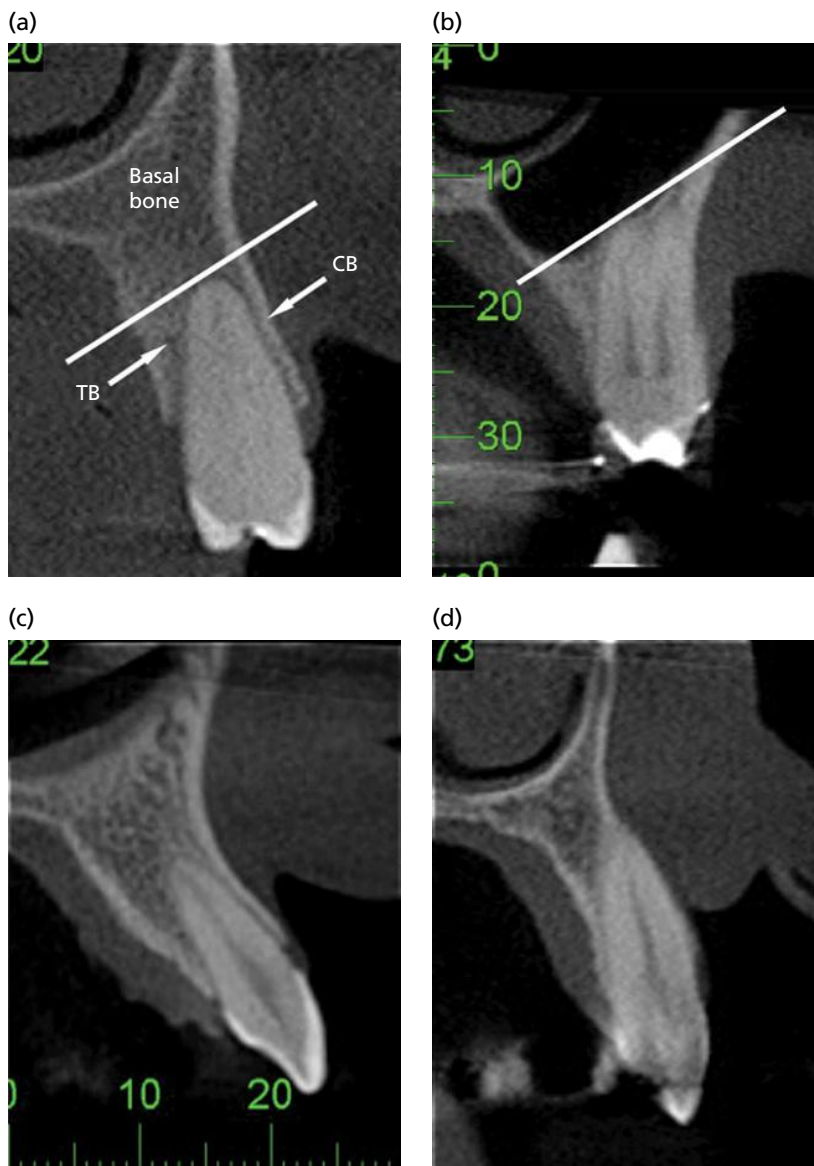


Fig. 3-10 (a) Cone-beam tomogram of the premolar region of the maxilla. The alveolar process is continuous with the voluminous basal bone of the maxilla. (CB, cortical bone plate; TB, trabecular bone.) (b) Cone-beam tomogram of the premolar region of the maxilla. Note that at this site the dimension of the basal bone is very small. (c) Tomogram of an anterior maxillary tooth with a “normal” direction of eruption. The incisor resides within the bony compartment of the alveolar process. (d) Tomogram of a canine tooth that erupted in a facial orientation. The facial (buccal) bone of the alveolar process is thin or even absent.

used by Amler only allowed the study of healing in the marginal portions of the empty socket, his findings are often referred to.

Amler stated that following tooth extraction, the first 24 hours are characterized by the formation of a *blood clot* in the socket. Within 2–3 days the blood clot is gradually replaced with *granulation tissue*. After 4–5 days, the *epithelium* from the margins of the soft tissue starts to proliferate to cover the granulation tissue in the socket. One week after extraction, the socket contains granulation tissue and *young connective tissue*, and *osteoid* formation is ongoing in the apical portion of the socket. After 3 weeks, the socket contains connective tissue and there are signs of mineralization of the osteoid. The *epithelium* covers the wound. After 6 weeks of healing, bone formation in the socket is pronounced and trabeculae of newly formed bone can be seen.

Amler's study was of short duration, so it could only evaluate events that took place in the marginal portion of the healing socket. His experimental data did not include the important later phase of socket healing that involves the processes of modeling and remodeling of the newly formed tissue in various parts of the alveolus. Thus, the tissue composition of the fully healed extraction site was not documented in the study.

In a later and longer-term study, Trombelli *et al.* (2008) examined socket healing in biopsies sampled during a 6-month period from human volunteers. They confirmed most of Amler's findings and reported that in the early healing phase (tissue modeling), the socket was filled with granulation tissue that was subsequently replaced with a provisional connective tissue and woven bone. In biopsies sampled in later phases of healing, it was observed that the process by which woven bone was replaced by lamellar bone and marrow, that is remodeling, was slow and exhibited great individual variation. In only a limited number of specimens representing 6 months of healing had woven bone been replaced with bone marrow and trabeculae of lamellar bone. It can be assumed therefore that tissue modeling following tooth extraction in humans is a rather rapid process, while the subsequent remodeling is slow and may take years to be completed.

The results from experiments using the dog model (Cardaropoli *et al.* 2003, Araújo & Lindhe 2005) will be used in this chapter to describe details of the various phases of socket healing, including processes of both modeling and remodeling. It should be remembered that healing of the post-extraction sites in these animal studies, including phases of modeling and remodeling, was a rapid process compared to socket healing in humans. Thus, the extraction socket was in most instances completely healed (filled with cancellous bone) after 2–3 months.

The model

Buccal and lingual full-thickness flaps are elevated and the distal roots of mandibular premolars extracted (Fig. 3-11a). The mucosal flaps are subsequently

replaced to provide soft tissue coverage of the fresh extraction wound (Fig. 3-11b). Healing of the experimental sites is monitored in biopsy specimens obtained at time intervals varying from 1 day to 6 months (Fig. 3-11c).

Overall pattern of socket healing

Figure 3-12 shows a mesiodistal section of a fresh extraction socket bordered by adjacent roots. The socket walls are continuous with the alveolar bone proper of the neighboring teeth. The tissue inside the interdental (inter-radicular) septa is made up of cancellous bone and includes trabeculae of lamellar bone within bone marrow.

The empty socket is first filled with blood and a *coagulum* (clot) forms (Fig. 3-13a). Inflammatory cells (polymorphonuclear leukocytes and monocytes/macrophages) migrate into the coagulum and start to phagocytose elements of necrotic tissue. The process of wound cleansing is initiated (Fig. 3-13b). Sprouts of newly formed vessels and mesenchymal cells (from the severed periodontal ligament) enter the coagulum and *granulation tissue* is formed. The granulation tissue is gradually replaced with *provisional connective tissue* (Fig. 3-13c) and subsequently immature bone (*woven bone*) is laid down (Fig. 3-13d). The hard tissue walls of the socket – the alveolar bone proper or the bundle bone – are gradually resorbed and the socket becomes filled with immature woven bone (Fig. 3-13e). The initial phase of the healing process (tissue modeling) is now complete. In subsequent phases, the woven bone in the socket will be gradually remodeled into lamellar bone and marrow (Fig. 3-13f–h).

Important events in socket healing

Blood clotting

Immediately after tooth extraction, blood from the severed vessels will fill the socket. Proteins derived from vessels and damaged cells initiate a series of events that lead to the formation of a fibrin network (Fig. 3-14). *Platelets* form aggregates and interact with the fibrin network to produce a *coagulum* (a blood clot) that effectively plugs the severed blood vessels and stops the bleeding. The blood clot acts as a physical matrix that directs cellular movements and it contains substances that are of importance for the forthcoming healing process. Thus, the clot contains substances (i.e. *growth factors*) that (1) influence mesenchymal cells and (2) enhance the activity of inflammatory cells. Such substances will thus induce and amplify the migration of various types of cells into the socket wound, as well as their proliferation, differentiation, and synthetic activity within the coagulum.

Although the blood clot is crucial in the initial phase of wound healing, its removal is mandatory to allow the formation of new tissue. Thus, within a few days after the tooth extraction, the blood clot will

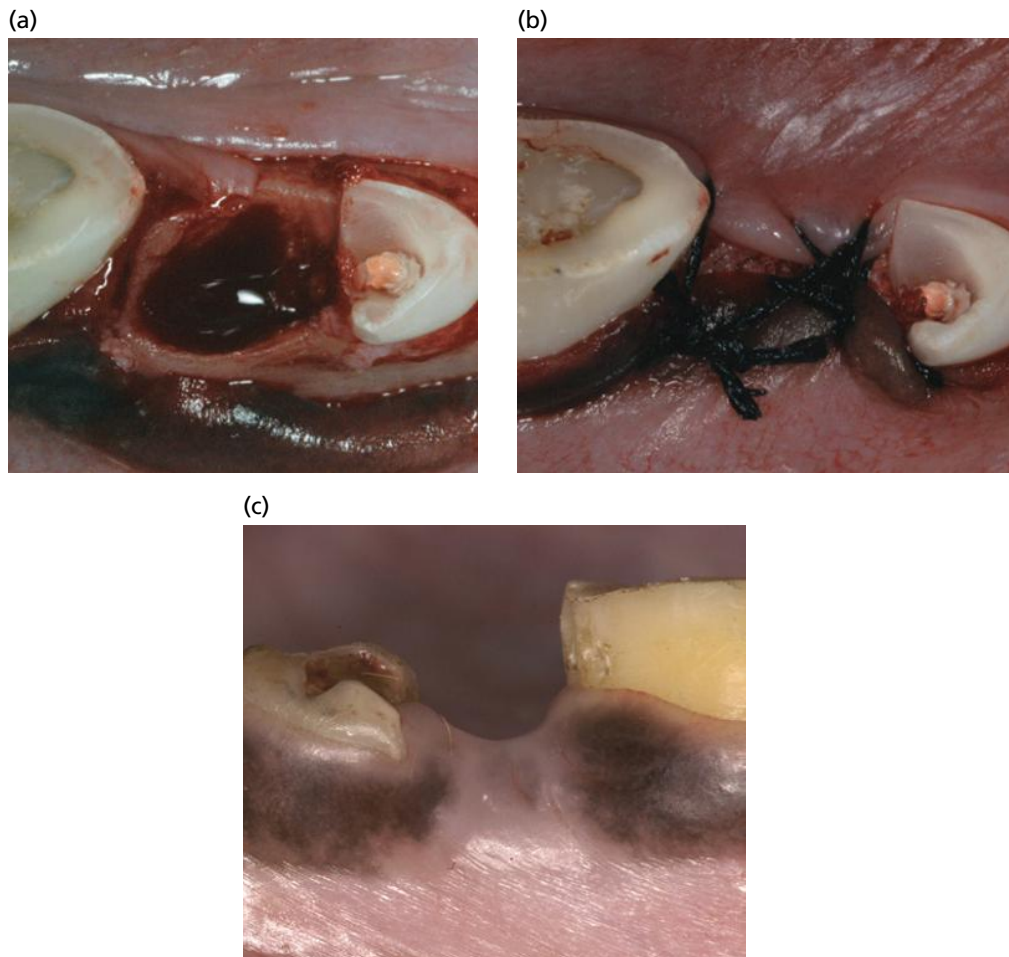


Fig. 3-11 (a) Photograph illustrating a mandibular premolar site (from a dog model) from which the distal root of the fourth premolar had been removed. (b) Mucosal, full-thickness flaps were replaced and sutured to close the entrance of the socket. (c) Site after 6 months of healing. Note the saddle-shaped outline (loss of tissue) of the alveolar crest region.



Fig. 3-12 Histologic section showing the mesiodistal aspect of a fresh extraction socket bordered by two neighboring roots. Note that the alveolar bone from the tooth sites is continuous with the walls of the empty socket. The interdental septum contains cancellous bone including trabeculae of lamellar bone and marrow.

start to break down, that is the process of “fibrinolysis” is initiated (Fig. 3-15).

Wound cleansing

Neutrophils and macrophages migrate into the wound, engulf bacteria and damaged tissue, and clean the site before the formation of new tissue can start. The neutrophils enter the wound early, while macrophages appear somewhat later. The macrophages are not only involved in the cleaning of the wound but they also release growth factors and cytokines that further promote the migration, proliferation, and differentiation of mesenchymal cells. Once the debris has been removed and the wound has been “sterilized”, the neutrophils undergo a programmed cell death (*apoptosis*) and are removed from the site through the action of macrophages. The macrophages subsequently withdraw from the wound.

Tissue formation

Sprouts of vascular structures (from the severed periodontal ligament) as well as mesenchymal, fibroblast-like cells (from the periodontal ligament and from adjacent bone marrow regions) enter the socket. The mesenchymal cells start to proliferate and deposit

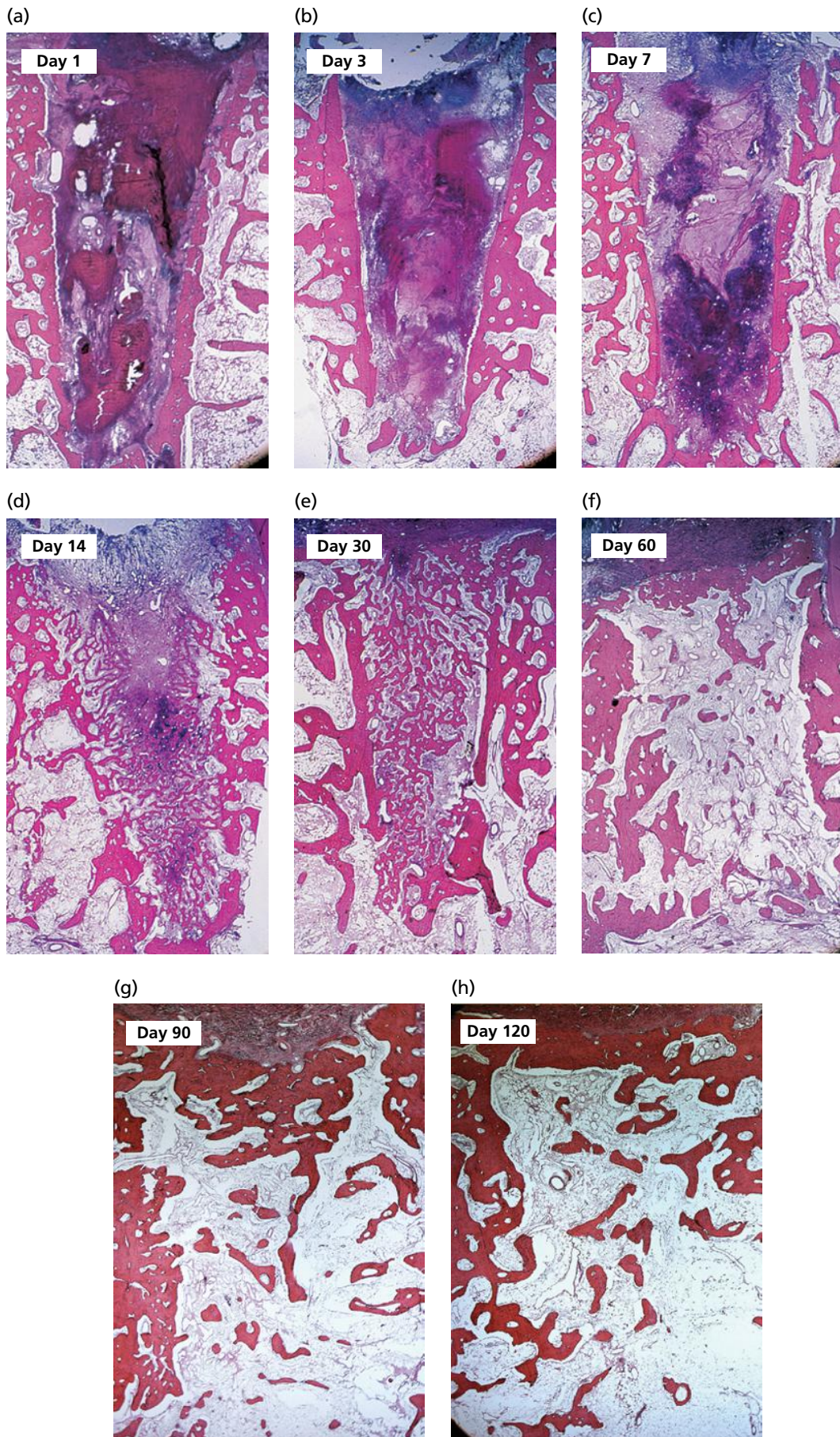


Fig. 3-13 (a–h) Overall pattern of bone formation in an extraction socket. For details see text.

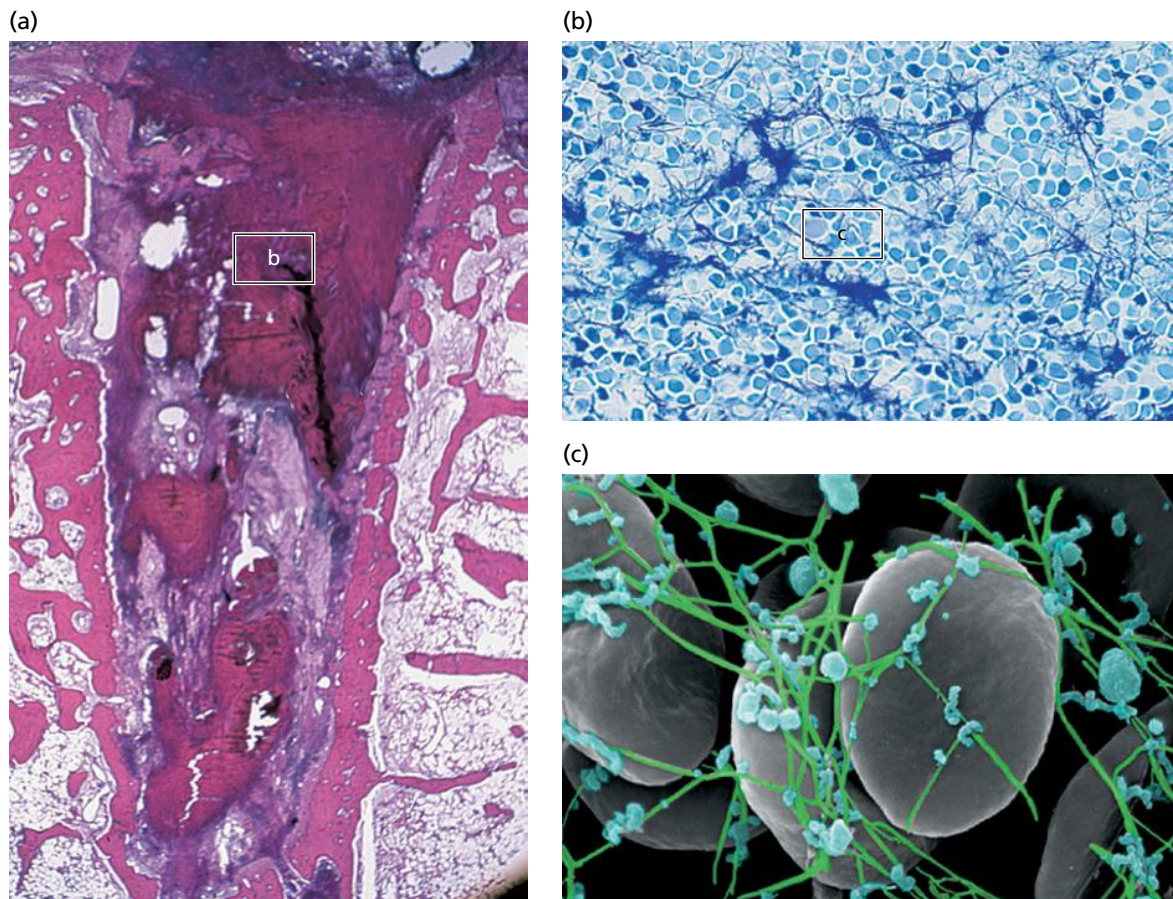


Fig. 3-14 Histologic section (mesiodistal aspect) representing 1 day of healing (a). The socket is occupied with a blood clot that contains large numbers of erythrocytes (b) entrapped in a fibrin network, as well as platelets [blue in (c)].

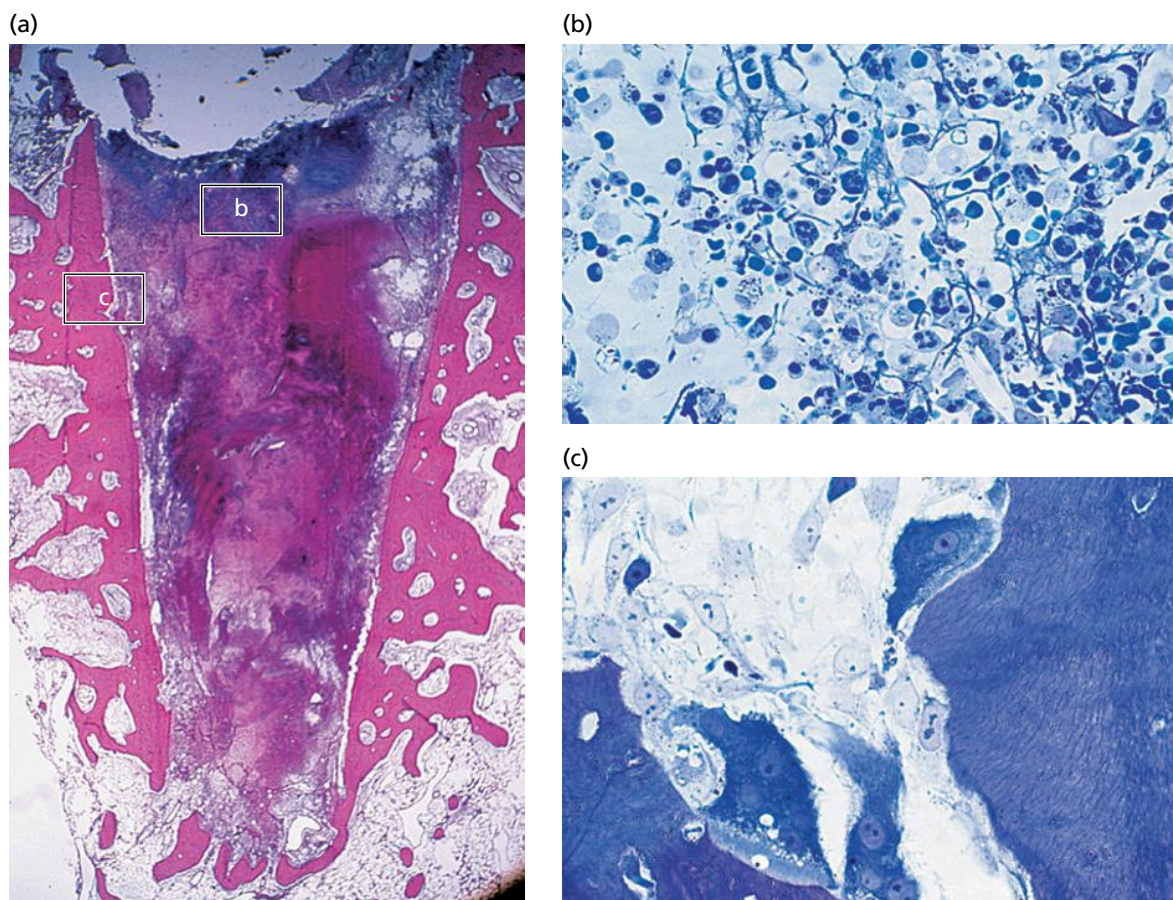


Fig. 3-15 (a) Histologic section (mesiodistal aspect) representing 3 days of healing. (b) Note the presence of neutrophils and macrophages that are engaged in wound cleansing and the breakdown of the blood clot. (c) Osteoclastic activity occurs on the surface of the old bone in the socket walls.

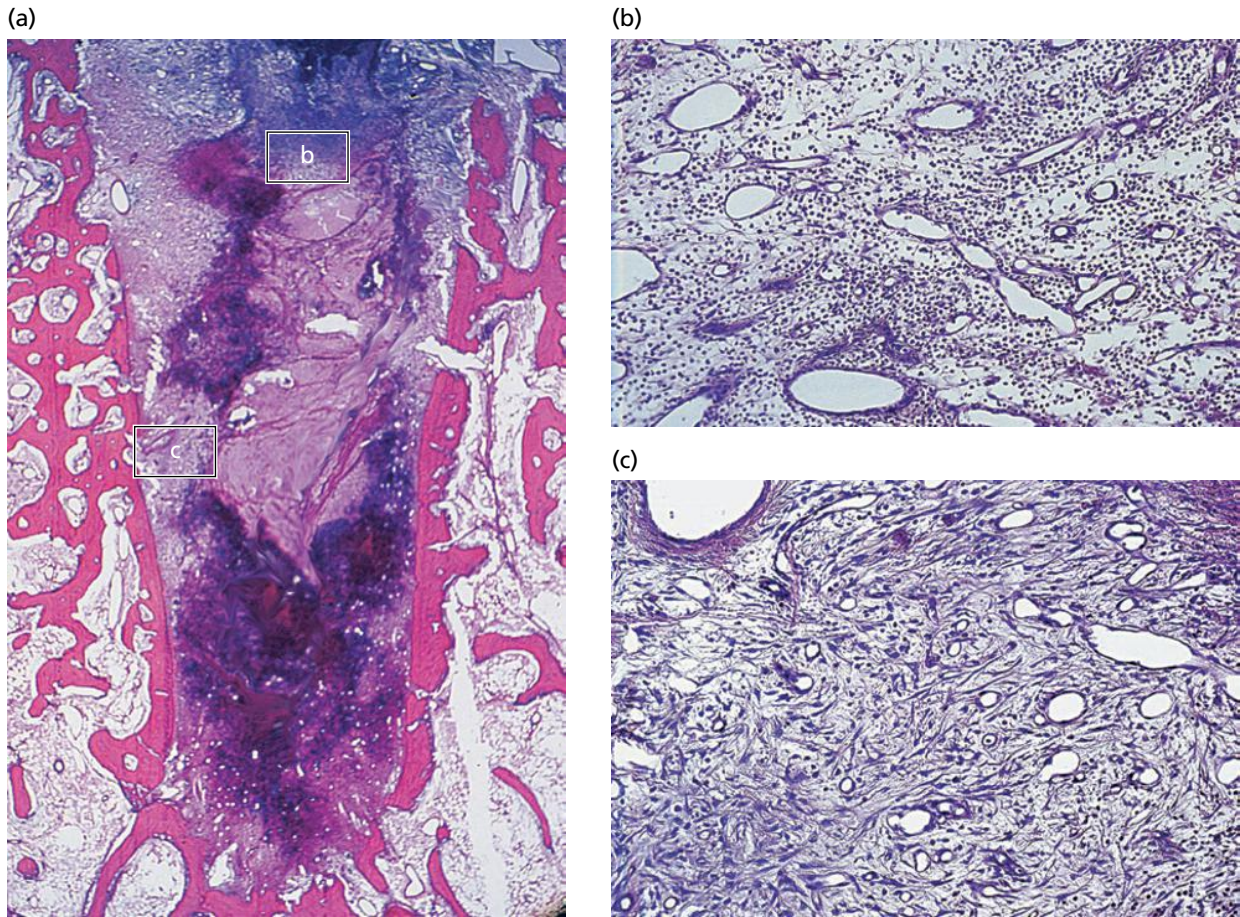


Fig. 3-16 (a) Histologic section (mesiodistal aspect) representing 7 days of healing. (b) Note the presence of a richly vascularized early granulation tissue with large numbers of inflammatory cells in the upper portion of the socket. (c) In more apical areas, a tissue including large numbers of fibroblast-like cells is present (late granulation tissue).

matrix components in an extracellular location (Fig. 3-16); *granulation tissue* will gradually replace the blood clot. This granulation tissue eventually contains macrophages and a large number of fibroblast-like cells, as well as numerous newly formed blood vessels. The fibroblast-like cells continue to (1) release growth factors, (2) proliferate, and (3) deposit a new extracellular matrix that guides the ingrowth of additional cells and allows the further differentiation of the tissue. The newly formed vessels provide the oxygen and nutrients that are needed for the increasing number of cells that occur in the new tissue. The intense synthesis of matrix components exhibited by the mesenchymal cells is called *fibroplasia*, while the formation of new vessels is called *angiogenesis*. A *provisional connective tissue* is established through the combination of fibroplasia and angiogenesis (Fig. 3-17).

The transition of the provisional connective tissue into bone tissue occurs along the vascular structures. Thus, osteoprogenitor cells (e.g. pericytes) migrate and gather in the vicinity of the vessels. They differentiate into osteoblasts that produce a matrix of collagen fibers, which takes on a woven pattern. The *osteoid* is formed. The process of mineralization is initiated within the osteoid. The osteoblasts continue to lay down osteoid and occasionally such cells are trapped in the matrix and become osteocytes. This newly formed bone is called *woven bone* (Figs. 3-17, 3-18).

The woven bone is the first type of The transition of the provisional connective tissue into bone tissue occurs along the vascular structures. Thus, osteoprogenitor cells (e.g. pericytes) migrate and gather in the vicinity of the vessels. They differentiate into osteoblasts that produce a matrix of collagen fibers, which takes on a woven pattern. The *osteoid* is formed. The process of mineralization is initiated within the osteoid. The osteoblasts continue to lay down osteoid and occasionally such cells are trapped in the matrix and become osteocytes. This newly formed bone is called *woven bone* (Figs. 3-17, 3-18).bone to be formed and is characterized by (1) its rapid deposition as finger-like projections along the route of vessels, (2) the poorly organized collagen matrix, (3) the large number of osteoblasts that are trapped in its mineralized matrix, and (4) its low load-bearing capacity. Trabeculae of woven bone are shaped around and encircle the vessel. The trabeculae become thicker through the deposition of additional woven bone. Cells (osteocytes) become entrapped in the bone tissue and the first set of osteons, the *primary osteons*, are organized. The woven bone is occasionally reinforced by the deposition of so-called *parallel-fibered bone* (collagen fibers organized not in a woven but in a concentric pattern).

It is important to realize that during this early phase of healing most of the bone tissue in the walls of the socket (the bundle bone) is removed.

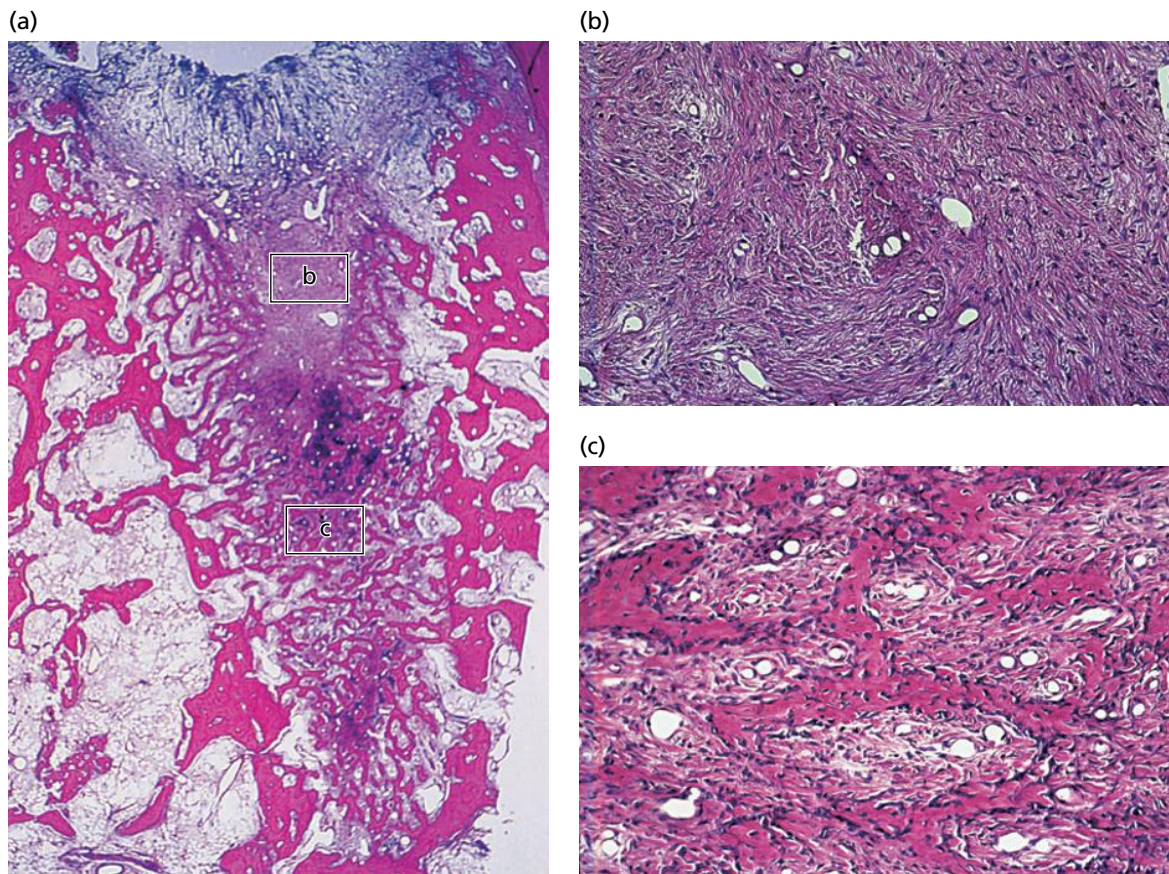


Fig. 3-17 (a) Histologic section (mesiodistal aspect) representing 14 days of healing. (b) In the marginal portion of the wound, a provisional connective tissue rich in fibroblast-like cells is present. (c) The formation of woven bone has at this time interval already begun in the apical and lateral regions of the socket.

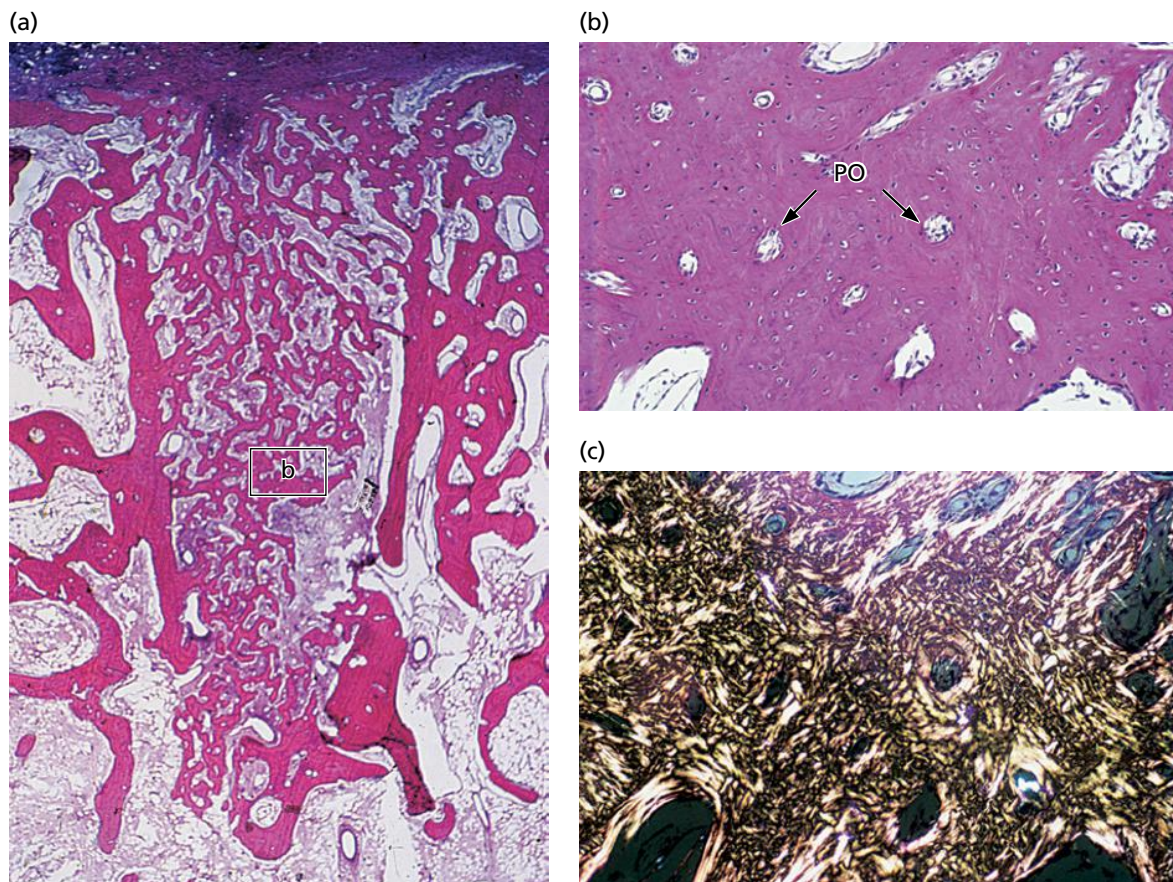


Fig. 3-18 (a) Histologic section (mesiodistal aspect) representing 30 days of healing. The socket is filled with woven bone. (b) Woven bone contains a large number of cells and primary osteons (PO). (c) The woven pattern of the collagen fibers of this type of bone is illustrated (polarized light).

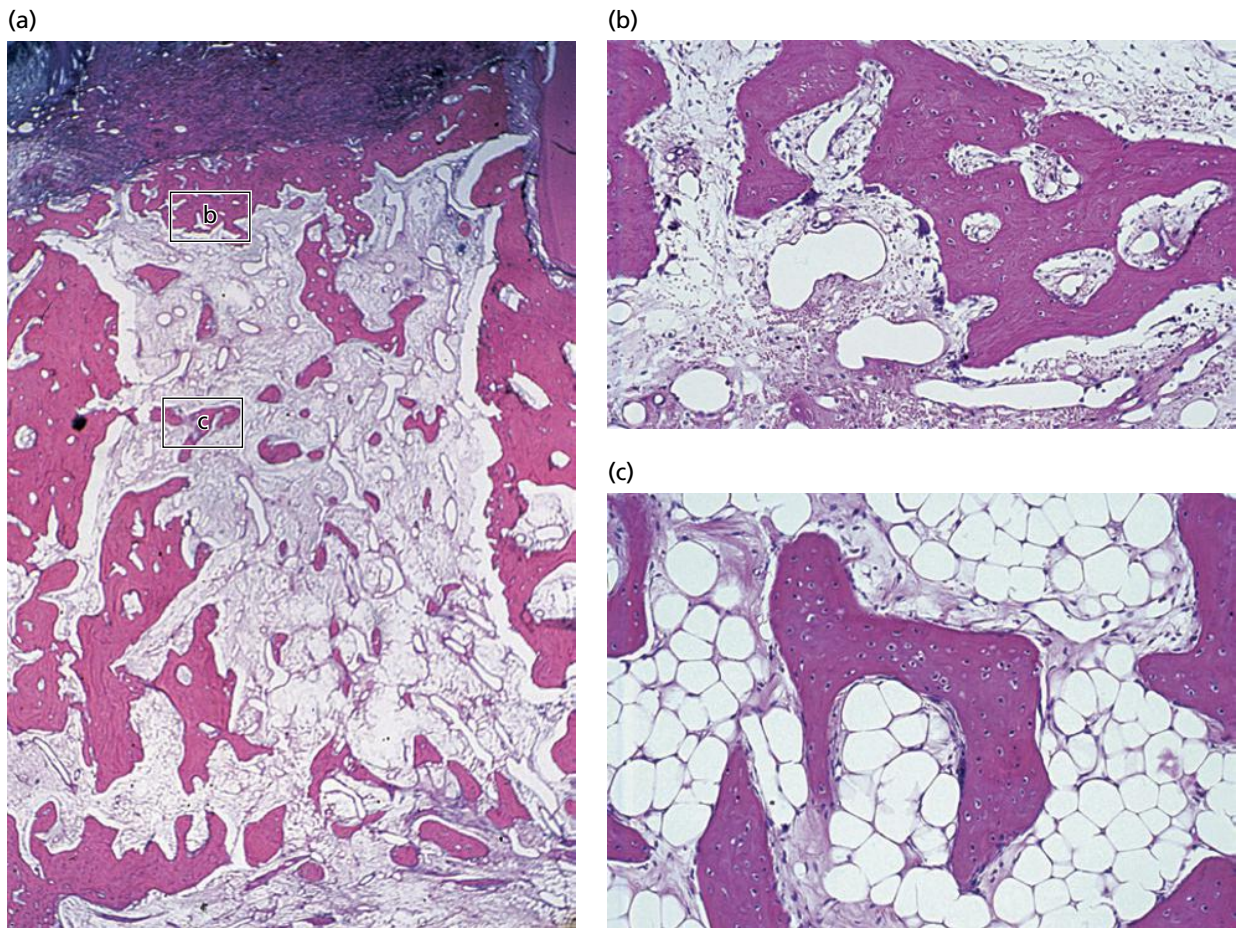


Fig. 3-19 (a) Histologic section (mesiodistal aspect) representing 60 days of healing. (b) A large portion of the woven bone has been replaced with bone marrow. (c) Note the presence of a large number of adipocytes residing in a tissue that still contains woven bone.

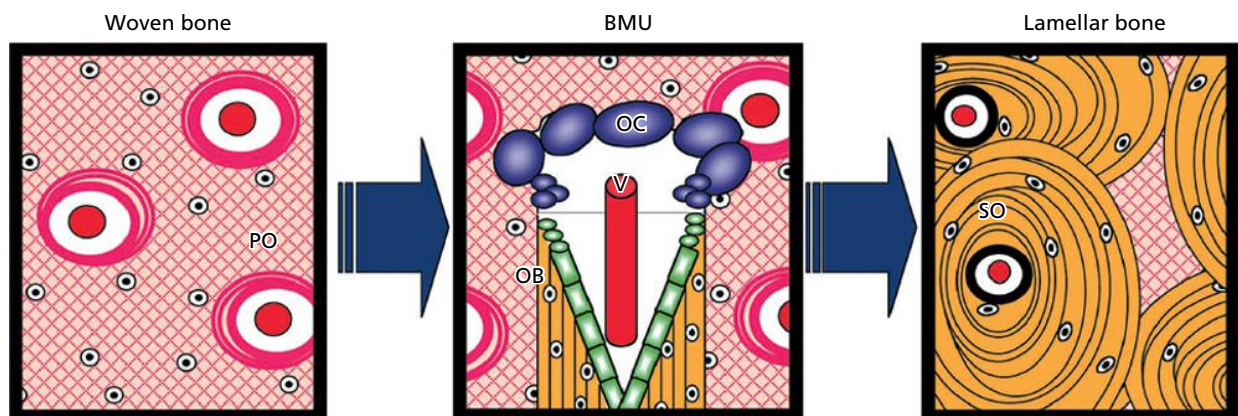


Fig. 3-20 Schematic drawing showing how woven bone is replaced by lamellar bone. Woven bone with primary osteons (PO) is substituted by lamellar bone in a process that involves the presence of bone multicellular units (BMU). The BMU contains osteoclasts (OC), as well as vascular structures (V) and osteoblasts (OB). Thus, the osteoblasts in the BMU produce bone tissue in a concentric fashion around the vessel, and lamellar bone with secondary osteons (SO) is formed.

Tissue modeling and remodeling

The initial bone formation in this dog model is a fast process. Within a few weeks, the entire extraction socket is filled with woven bone or, as this tissue is also called, *primary bone spongiosa*. The woven bone offers (1) a stable scaffold, (2) a solid surface, (3) a source of osteoprogenitor cells, and (4) an ample blood supply for cell function and matrix mineralization.

The woven bone with its primary osteons is gradually replaced with lamellar bone and bone marrow

(Fig. 3-19). In this process, the primary osteons are replaced with *secondary osteons*. The woven bone is first resorbed to a certain level. The level of the resorption front will establish a so-called *reversal line*, which is also the level from which new bone with secondary osteons will form (Fig. 3-20). Although this remodeling may start early during socket healing, it will take several months until all woven bone in the extraction socket has been replaced with lamellar bone and marrow.

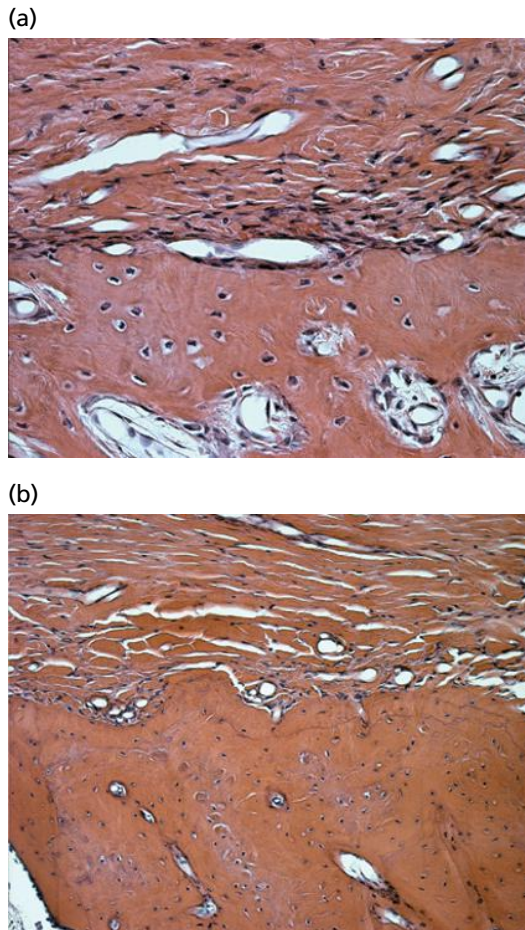


Fig. 3-21 Histologic sections (mesiodistal aspect) describing the hard tissue that has formed at the entrance of a healing extraction socket and the process of corticalization. (a) Woven bone with primary osteons occupies the socket entrance after 60 days of healing. (b) After 180 days, the woven bone has mainly been replaced with lamellar bone.

An important part of socket healing involves the formation of a *hard tissue cap* that will close the marginal entrance to the socket. This cap is initially comprised of woven bone (Fig. 3-21a), but is subsequently remodeled and replaced with lamellar bone that becomes continuous with the cortical plate at the periphery of the edentulous site (Fig. 3-21b). This process is called corticalization.

The wound is now healed, but the tissues in the site will continue to adapt to functional demands. Since there is no stress from forces elicited during mastication and other occlusal contacts, there is no demand on the mineralized bone in the areas previously occupied by the tooth. Thus, in this model the socket apical of the hard tissue cap will remodel mainly into marrow.

Extra-alveolar processes

In an experiment using the dog model (Araújo & Lindhe, 2005), alterations in the profile of the edentulous ridge that occurred following tooth extraction were carefully examined. In this study the third and fourth mandibular premolars were hemi-sectioned. Buccal and lingual full-thickness flaps were raised; the distal roots were carefully removed. The flaps were replaced and sutured to cover the fresh extraction socket. Biopsy specimens, including an individual extraction socket and adjacent roots, were obtained after 1, 2, 4, and 8 weeks of healing. The blocks were sectioned in the *buccolingual* plane.

- *1 week after tooth extraction* (Fig. 3-22). At this interval the socket is occupied by a coagulum. Furthermore, a large number of osteoclasts can be seen on the outside as well as on the inside of the buccal and

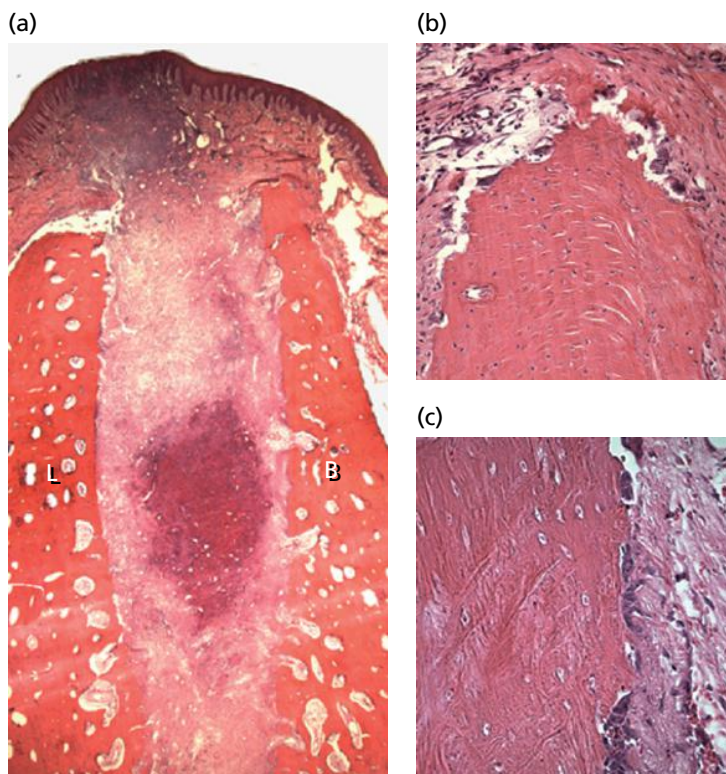


Fig. 3-22 (a) Histologic section (buccolingual aspect) of the socket after 1 week of healing. Note the presence of a large number of osteoclasts on the crestal portion (b) and inner portion (c) of the buccal wall. (B, buccal bone; L, lingual bone.)

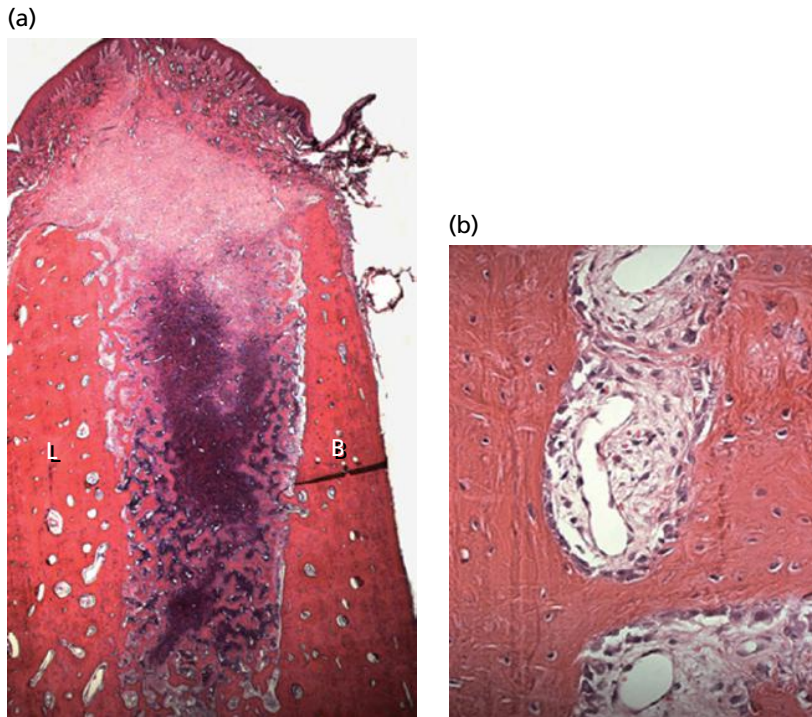


Fig. 3-23 (a) Histologic section (buccolingual aspect) of the socket after 2 weeks of healing. (b) Note that the bundle bone in the lingual aspect of the socket is being replaced with woven bone. (B, buccal bone; L, lingual bone.)

lingual bone walls. The presence of osteoclasts on the inner surface of the socket walls indicates that the bundle bone is being resorbed.

- *2 weeks after tooth extraction* (Fig. 3-23). Newly formed immature bone (woven bone) resides in the apical and lateral parts of the socket, while more central and marginal portions are occupied by a provisional connective tissue. In the marginal and outer portions of the socket walls, numerous osteoclasts can be seen. In several parts of the socket walls the bundle bone has been replaced with woven bone.
- *4 weeks after tooth extraction* (Fig. 3-24). The entire socket is occupied with woven bone at this stage of healing. Large numbers of osteoclasts are present in the outer and marginal portions of the hard tissue walls. Osteoclasts also line the trabeculae of woven bone present in the central and lateral aspects of the socket. In other words, the newly formed woven bone is being replaced with a more mature type of bone.
- *8 weeks after tooth extraction* (Fig. 3-25). A layer of cortical bone covers the entrance to the extraction site. Corticalization has occurred. The woven bone that was present in the socket at the 4-week interval is replaced with bone marrow and some trabeculae of lamellar bone in the 8-week specimens. On the outside and on the top of the buccal and lingual bone wall there are signs of ongoing hard tissue resorption. The crest of the buccal bone wall is located apical of its lingual counterpart.

The relative change in the location of the crest of the buccal and lingual bone walls that took place during the 8 weeks of healing is illustrated in Fig. 3-26. While the level of the margin of the lingual wall remained reasonably unchanged, the margin of the buccal wall

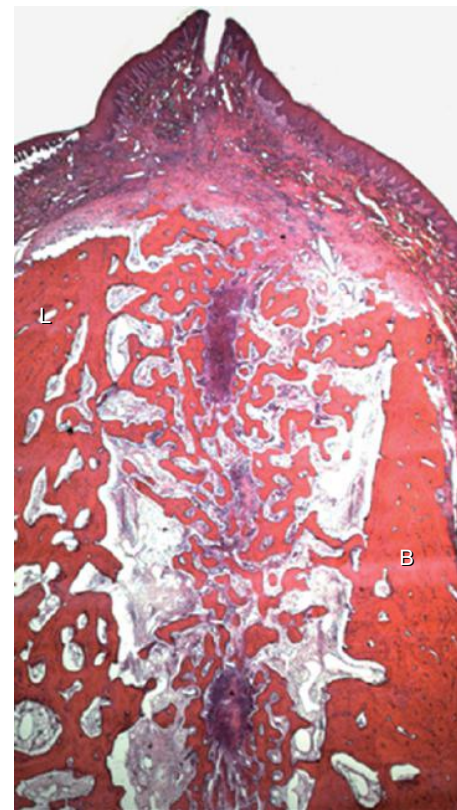


Fig. 3-24 Histologic section (buccolingual aspect) of the socket after 4 weeks of healing. The extraction socket is filled with woven bone. On the top of the buccal wall, the old bone in the crest region is being resorbed and replaced with either connective tissue or woven bone. (B, buccal bone; L, lingual bone.)

shifted several millimeters in an apical direction. The reason why more bone loss occurred in the buccal than in the lingual wall during socket healing in this animal model is not completely understood.

Prior to tooth extraction, the marginal 1–2 mm of the crest of the thin buccal bone wall was occupied by bundle bone. Only a minor fraction of the crest of the wider lingual wall contained bundle bone. Bundle bone, as stated above, is a tooth-dependent tissue and will gradually disappear after tooth extraction. Thus, since there is relatively more bundle

bone in the crest region of the buccal than of the lingual wall, hard tissue loss may become most pronounced in the buccal wall.

Topography of the edentulous ridge: Summary

As described previously in this chapter, the processes of modeling and remodeling that occur following tooth extraction (loss) result in resorption of the various components of the previous alveolar process. The amount of tissue loss that occurs in these processes varies considerably from subject to subject and from site to site in the same individual (Figs. 3-27, 3-28).

As a rule, the resorption of the buccal bone wall is more pronounced than the resorption of the lingual/palatal wall and hence the center of the ridge will move in a lingual/palatal direction. In the extreme case, the entire alveolar process may be lost following tooth removal and then only the basal bone of the mandible and the maxilla may remain to constitute the ridge.

The outer (cortical) walls of the remaining portion of the alveolar ridge (basal bone and residues of the alveolar process) are comprised of lamellar bone. The cortical plates of the ridge often enclose the cancellous bone that harbors trabeculae of lamellar bone and marrow (Fig. 3-29). The bone marrow contains numerous vascular structures as well as adipocytes and pluripotent mesenchymal cells.

Depending on factors such as the type of jaw (maxilla or mandible), location (anterior, posterior) in the jaw, depth of the buccal and lingual vestibule, and amount of hard tissue resorption, the edentulous ridge may be lined with either masticatory, keratinized mucosa, or lining, non-keratinized mucosa.

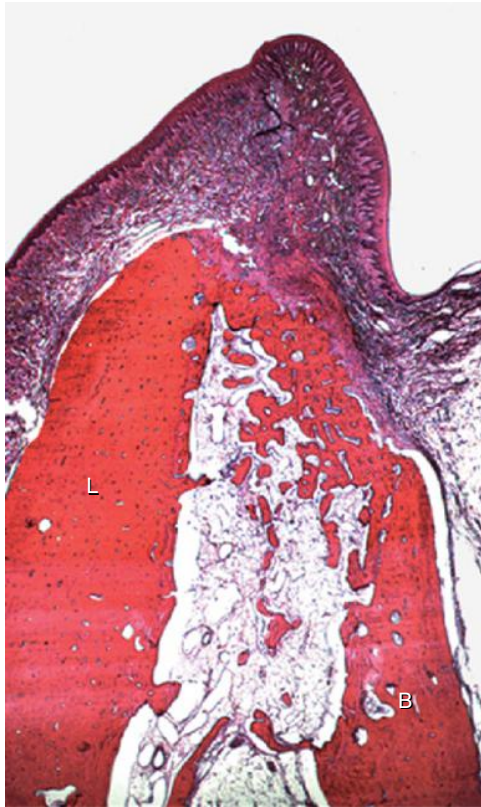


Fig. 3-25 Histologic section (buccolingual aspect) of the socket after 8 weeks of healing. The entrance of the socket is sealed with a cap of newly formed mineralized bone. Note that the crest of the buccal wall is located apical of the crest of the lingual wall. (B, buccal bone; L, lingual bone.)

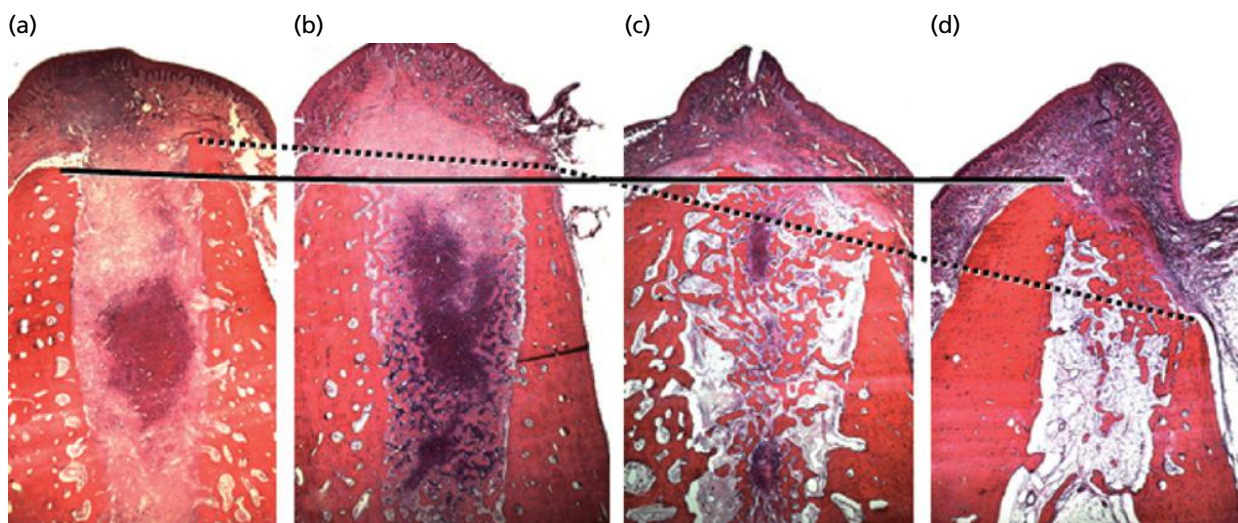


Fig. 3-26 Histologic sections (buccolingual aspects) showing the profile of the edentulous region in the dog after (a) 1, (b) 2, (c) 4, and (d) 8 weeks of healing following tooth extraction. While the marginal level of the lingual wall was maintained during the process of healing (solid line), the crest of the buccal wall was displaced >2 mm in the apical direction (dotted line).

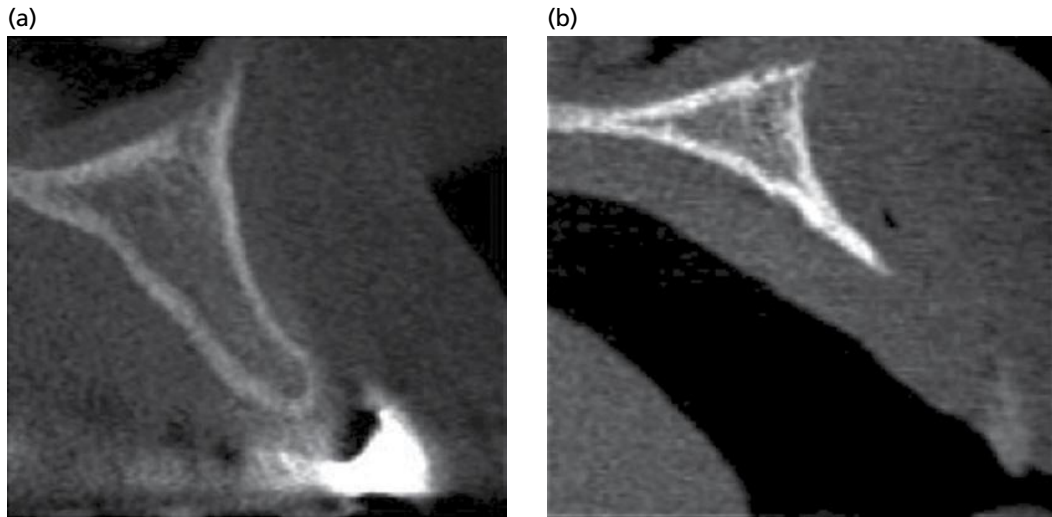


Fig. 3-27 Cone-beam computed tomograms that illustrate edentulous incisor sites of the maxilla with (a) large amounts of remaining hard tissue (cortical bone as well as trabecular bone) and (b) minute remnants of ridge tissue (only cortical bone).

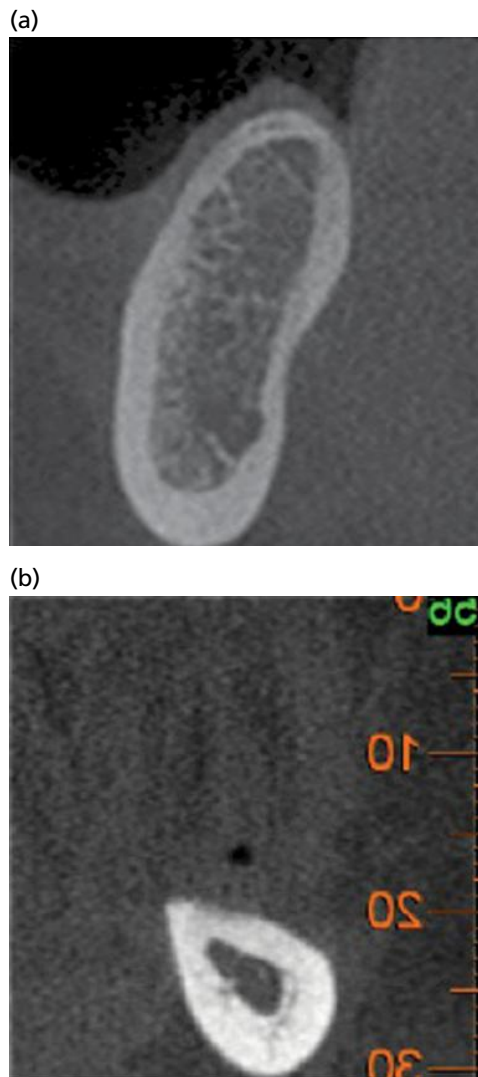


Fig. 3-28 Cone-beam computed tomograms illustrating edentulous regions of the first molar region of the mandible. (a) Remaining bone of the ridge is voluminous, is lined by dense cortical bone, and harbors large amounts of trabecular bone. (b) In this edentulous site, the entire alveolar process is lost and only the tissue of the corpus mandibulae remains.

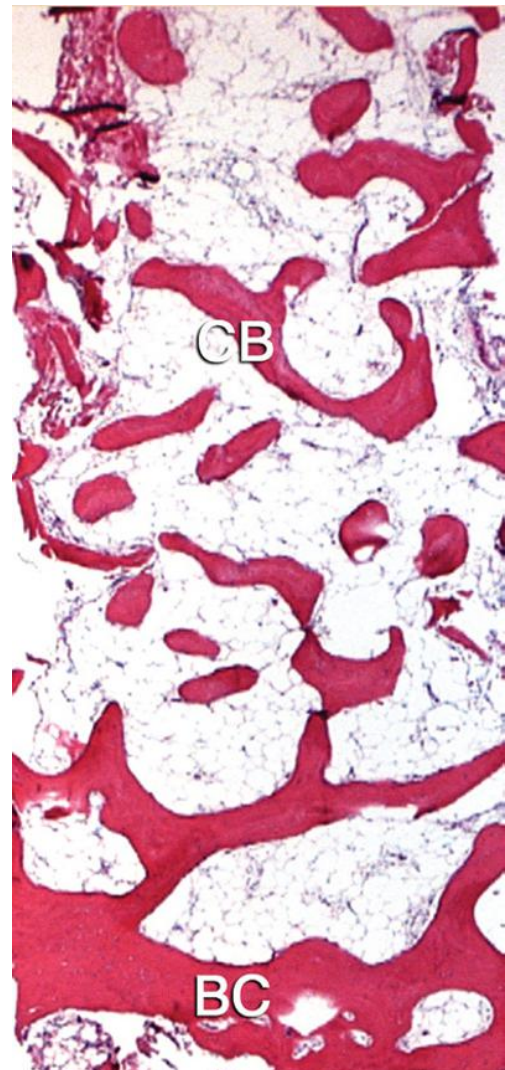


Fig. 3-29 Histologic section representing an edentulous maxilla. The biopsy was obtained >6 months post extraction. The marginal portion of the tissue (the bone crest [BC]) is comprised of dense lamellar bone, while more central portions harbor the cancellous bone (CB).

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